

Formal Grammars

The Chomsky Hierarchy

Grammars

Grammars express languages

Example: the English language grammar

$$\langle sentence \rangle \rightarrow \langle noun_phrase \rangle \langle predicate \rangle$$
$$\langle noun_phrase \rangle \rightarrow \langle article \rangle \langle noun \rangle$$
$$\langle predicate \rangle \rightarrow \langle verb \rangle$$

$\langle \textit{article} \rangle \rightarrow a$

$\langle \textit{article} \rangle \rightarrow \textit{the}$

$\langle \textit{noun} \rangle \rightarrow \textit{cat}$

$\langle \textit{noun} \rangle \rightarrow \textit{dog}$

$\langle \textit{verb} \rangle \rightarrow \textit{runs}$

$\langle \textit{verb} \rangle \rightarrow \textit{sleeps}$

Derivation of string "the dog walks":

$\langle sentence \rangle \Rightarrow \langle noun_phrase \rangle \langle predicate \rangle$
 $\Rightarrow \langle noun_phrase \rangle \langle verb \rangle$
 $\Rightarrow \langle article \rangle \langle noun \rangle \langle verb \rangle$
 $\Rightarrow the \langle noun \rangle \langle verb \rangle$
 $\Rightarrow the \ dog \langle verb \rangle$
 $\Rightarrow the \ dog \ sleeps$

Derivation of string "a cat runs":

$\langle sentence \rangle \Rightarrow \langle noun_phrase \rangle \langle predicate \rangle$
 $\Rightarrow \langle noun_phrase \rangle \langle verb \rangle$
 $\Rightarrow \langle article \rangle \langle noun \rangle \langle verb \rangle$
 $\Rightarrow a \langle noun \rangle \langle verb \rangle$
 $\Rightarrow a \text{ cat } \langle verb \rangle$
 $\Rightarrow a \text{ cat runs}$

Language of the grammar:

$$L = \{ \text{"a cat runs"}, \\ \text{"a cat sleeps"}, \\ \text{"the cat runs"}, \\ \text{"the cat sleeps"}, \\ \text{"a dog runs"}, \\ \text{"a dog sleeps"}, \\ \text{"the dog runs"}, \\ \text{"the dog sleeps"} \}$$

Productions

Sequence of
Terminals (symbols)

$\langle \textit{noun} \rangle \rightarrow \textit{cat}$

$\langle \textit{sentence} \rangle \rightarrow \langle \textit{noun_phrase} \rangle \langle \textit{predicate} \rangle$

Variables

Sequence of Variables

Another Example

Sequence of
terminals and variables

Grammar:

$$S \rightarrow \overbrace{aSb}$$

$$S \rightarrow \lambda$$

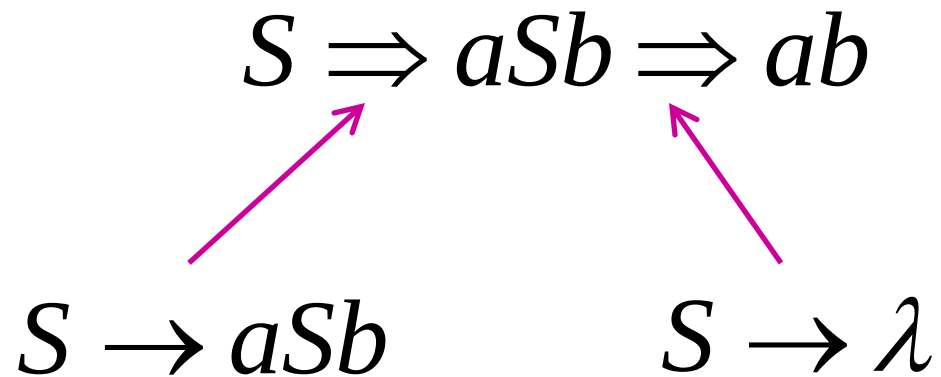
Variable

The right side
may be λ

Grammar: $S \rightarrow aSb$

$S \rightarrow \lambda$

Derivation of string ab :



Grammar: $S \rightarrow aSb$

$S \rightarrow \lambda$

Derivation of string $aabb$:

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$



$S \rightarrow aSb$



$S \rightarrow \lambda$

Grammar: $S \rightarrow aSb$

$S \rightarrow \lambda$

Other derivations:

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb \Rightarrow aaabbbb$

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb$
 $\Rightarrow aaaaSbbbb \Rightarrow aaabbbbb$

Grammar: $S \rightarrow aSb$

$$S \rightarrow \lambda$$

Language of the grammar:

$$L = \{a^n b^n : n \geq 0\}$$

A Convenient Notation

We write: $S \xRightarrow{*} aaabbb$

for zero or more derivation steps

Instead of:

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaasbbb \Rightarrow aaabbbb$

In general we write: $w_1 \xRightarrow{*} w_n$

If: $w_1 \Rightarrow w_2 \Rightarrow w_3 \Rightarrow \cdots \Rightarrow w_n$

in zero or more derivation steps

Trivially: $w \xRightarrow{*} w$

Example Grammar

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

Possible Derivations

$$\begin{array}{c} * \\ S \Rightarrow \lambda \end{array}$$

$$\begin{array}{c} * \\ S \Rightarrow ab \end{array}$$

$$\begin{array}{c} * \\ S \Rightarrow aaabbb \end{array}$$

$$S \xRightarrow{*} aaSbb \xRightarrow{*} aaaaaaSbbbb b$$

Another convenient notation:

$$\begin{array}{l} S \rightarrow aSb \\ S \rightarrow \lambda \end{array} \quad \longrightarrow \quad S \rightarrow aSb \mid \lambda$$

$$\begin{array}{l} \langle \textit{article} \rangle \rightarrow a \\ \langle \textit{article} \rangle \rightarrow \textit{the} \end{array} \quad \longrightarrow \quad \langle \textit{article} \rangle \rightarrow a \mid \textit{the}$$

Formal Definition

Grammar: $G = (V, T, S, P)$

Set of
Variables

(nonterminal)

Set of
terminal
symbols

Start
variable

Set of
productions

$$P = \{ \alpha \rightarrow \beta \mid \alpha \in (V \cup T)^+, \beta \in (V \cup T)^*, \alpha \text{ has at least a variable} \}$$

Example of a formal grammar:

$$G : A \rightarrow aABC \mid abC$$

$$CB \rightarrow BC$$

$$bB \rightarrow bb$$

$$bC \rightarrow bc$$

$$cC \rightarrow cc$$

Contracts:

$$V = \{A, B, C\}$$

$$T = \{a, b, c\}$$

$$S = A$$

Derivation of string abc : $A \Rightarrow abC \Rightarrow abc$

Derivation of string $aabbcc$:

$$A \Rightarrow aABC \Rightarrow aabCBC \Rightarrow aabBCC \Rightarrow aabbCC \Rightarrow aabbcC \Rightarrow aabbcc$$

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Derivation of string $aabbcc$:

$$A \Rightarrow aABC \Rightarrow aabCBC \Rightarrow aabBCC \Rightarrow aabbCC \Rightarrow aabbcC \Rightarrow aabbcc$$

The Language is $L(G) = \{a^n b^n c^n \mid n \geq 1\}$

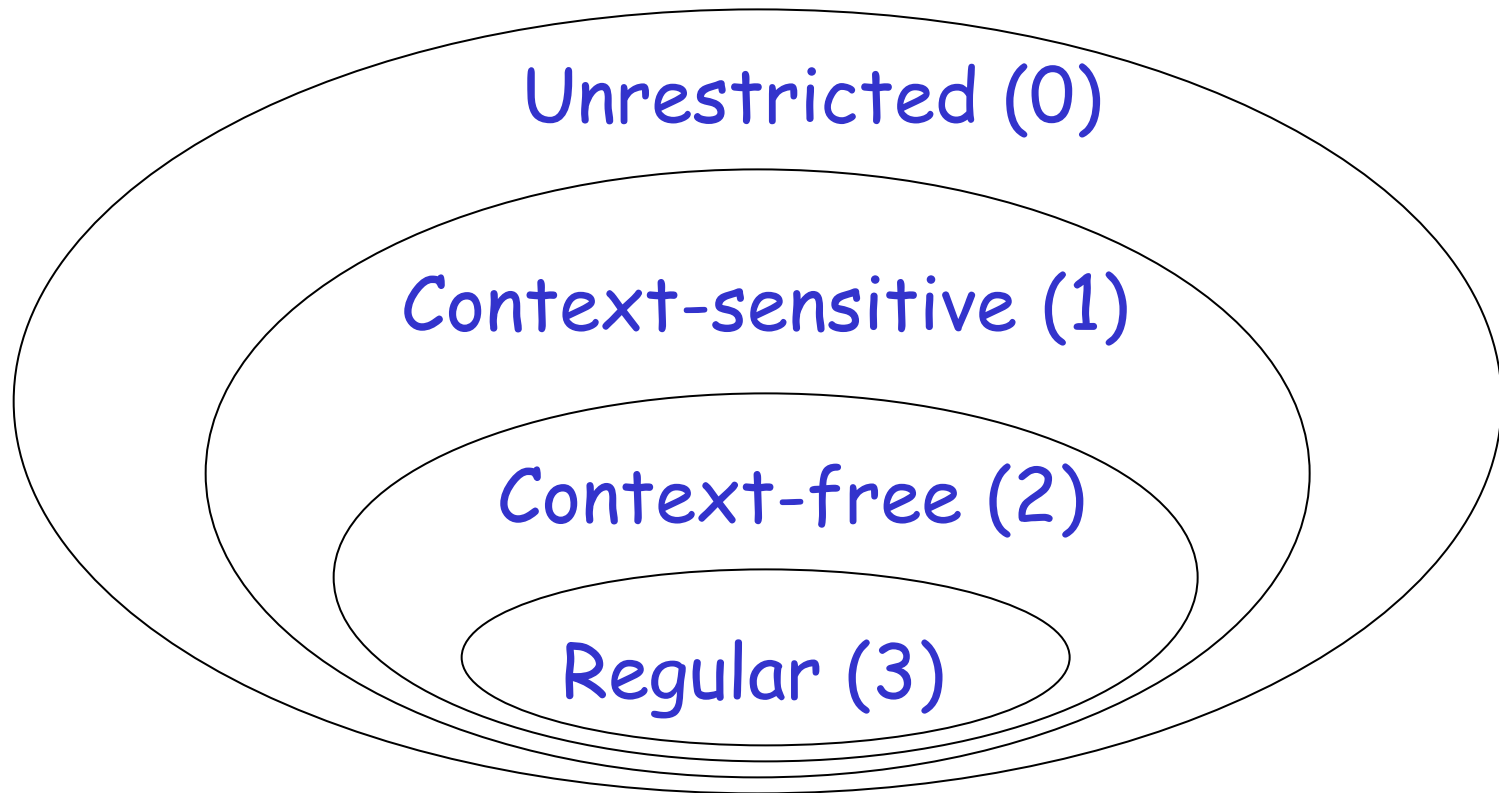
Language of a Grammar:

For a grammar G with start variable S

$$L(G) = \{w : S \xRightarrow{*} w, \quad w \in T^*\}$$

String of terminals or λ

The Chomsky Hierarchy (Grammars and Languages)

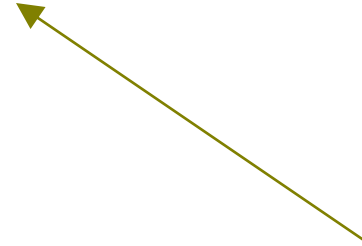
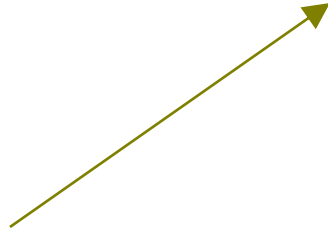


We say that a language L is of **type i** iff its **(most restricted)** grammar is of **type i** .

Unrestricted Grammars (Type 0):

Productions

$u \rightarrow v$



String of variables
and terminals

String of variables
and terminals

$$P = \{u \rightarrow v \mid u \in (V \cup T)^+, v \in (V \cup T)^*, u \text{ has at least a variable}\}$$

Example of an unrestricted grammar:

$$A \rightarrow aABC \mid abC$$

$$CB \rightarrow BC$$

$$bB \rightarrow bb$$

$$bC \rightarrow b$$

Derivation of string ab : $A \Rightarrow abC \Rightarrow ab$

Derivation of string $aabb$:

$$A \Rightarrow aABC \Rightarrow aabCBC \Rightarrow aabBCC \Rightarrow aabbCC \Rightarrow aabbC \Rightarrow aabb$$

The Language is $L(G) = \{a^n b^n \mid n \geq 1\}$

Example unrestricted grammar:

$$S \rightarrow aBc$$

$$aB \rightarrow cA$$

$$Ac \rightarrow d$$

Context-Sensitive Grammars:

Productions

$$u \rightarrow v$$


String of variables
and terminals

String of variables
and terminals

and: $|u| \leq |v|$

The language $L(G) = \{a^n b^n c^n \mid n \geq 1\}$
is context-sensitive:

$$S \rightarrow abc \mid aAbc$$

$$Ab \rightarrow bA$$

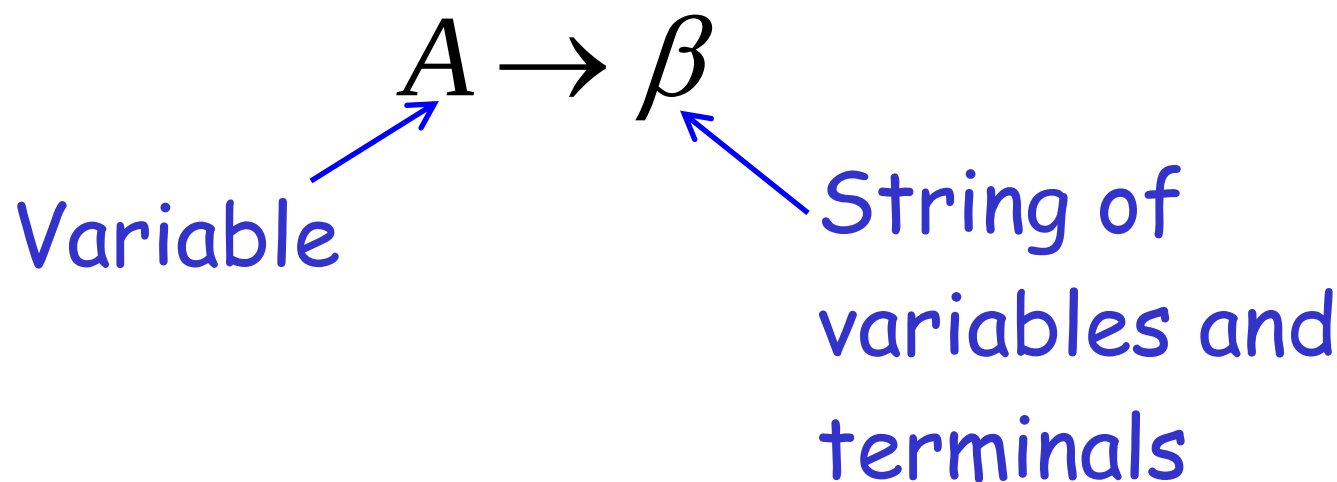
$$Ac \rightarrow Bbcc$$

$$bB \rightarrow Bb$$

$$aB \rightarrow aa \mid aaA$$

Context-Free Grammar: $G = (V, T, S, P)$

All productions in P are of the form



$$P = \{ A \rightarrow \beta \mid A \in V, \beta \in (V \cup T)^* \}$$

Example of Context-Free Grammar

$$S \rightarrow aSb \mid \lambda$$

productions

$$P = \{S \rightarrow aSb, S \rightarrow \lambda\}$$

$$G = (V, T, S, P)$$

$V = \{S\}$
variables

$T = \{a, b\}$
terminals

start variable

Language of a Grammar:

For a grammar G with start variable S

$$L(G) = \{w : S \Rightarrow^* w, \quad w \in T^*\}$$

String of terminals or λ

Example:

context-free grammar $G : \boxed{S \rightarrow aSb \mid \lambda}$

$$L(G) = \{a^n b^n : n \geq 0\}$$

Since, there is derivation

$$S \xRightarrow{*} a^n b^n \quad \text{for any } n \geq 0$$

Example:

$$L = \{a^n b^n : n \geq 0\}$$

is a context-free language

since context-free grammar G :

$$S \rightarrow aSb \mid \lambda$$

generates $L(G) = L$

Another Example

Context-free grammar G :

$$S \rightarrow aSa \mid bSb \mid \lambda$$

Example derivations:

$$S \Rightarrow aSa \Rightarrow abSba \Rightarrow abba$$

$$S \Rightarrow aSa \Rightarrow abSba \Rightarrow abaSaba \Rightarrow abaaba$$

$$L(G) = \{ww^R : w \in \{a,b\}^*\}$$

Palindromes of even length

Another Example

Context-free grammar G :

$$S \rightarrow aSb \mid SS \mid \lambda$$

Example derivations:

$$S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow ab$$

$$S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow abaSb \Rightarrow abab$$

$$L(G) = \{w : n_a(w) = n_b(w),$$

$$\text{and } n_a(v) \geq n_b(v)$$

in any prefix v }

Describes
matched

parentheses:

$() ((())) (())$ $a = (, \quad b =)$

(Right-Linear) Regular Grammars:

Productions:

$$A \rightarrow xB \mid x \quad A, B \in V \quad x \in T^*$$

(Left-Linear) Regular Grammars:

Productions:

$$A \rightarrow Bx \mid x \quad A, B \in V \quad x \in T^*$$

Regular Grammars:

all productions are right-linear **XOR**
all productions are left-linear.

Examples of Regular Grammar

$$G_1 : \quad A \rightarrow 1B \mid 1 \quad B \rightarrow 0A$$

$$G_2 : \quad A \rightarrow B1 \mid 1 \quad B \rightarrow A0$$

$$L(G_1) = \{1(01)^n \mid n \geq 0\} \equiv 1(01)^*$$

$$L(G_2) = \{(10)^n 1 \mid n \geq 0\} \equiv (10)^* 1$$

$$L(G_1) = L(G_2)$$

Examples of Regular Grammar

$$G_3 : A \rightarrow abA \mid a$$

$$G_4 : A \rightarrow Bab \quad B \rightarrow Bab \mid C \quad C \rightarrow a$$

$$L(G_3) = \{(ab)^n a \mid n \geq 0\} \equiv (ab)^* a$$

$$L(G_4) = \{aab(ab)^n \mid n \geq 0\} \equiv aab(ab)^*$$

(Right-Linear) Regular Grammars:

$$A \rightarrow xB \mid x \quad A, B \in V \quad x \in T^*$$

(Left-Linear) Regular Grammars:

$$A \rightarrow Bx \mid x \quad A, B \in V \quad x \in T^*$$

Two Main Theorems:

1) A language L is regular iff there exist a right-linear grammar G such that $L = L(G)$.

Proof:??

2) A language L is regular iff there exist a left-linear grammar G such that $L = L(G)$.

Proof:??

Corollary: for every right-linear grammar G_1 there is a left-linear grammar G_2 such that $L(G_1) = L(G_2)$ and vice versa.